

SRI AMAN SIMANGGAN, Malaysia

WMO: 960011

Lat: 1.217N

Lon: 111.450E

Elev: 36

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	71.4	71.9	69.5	108.6	74.7	70.0	110.3	74.8	10.0	82.6	9.1	82.6	1.6	110	0.280

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	16.0	95.0	81.1	93.6	81.1	93.1	81.1	83.4	91.6	82.7	90.8	82.0	90.1	3.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.9	160.9	89.0	80.6	159.2	88.7	79.2	151.9	87.1	47.1	88.6	46.4	91.2	45.8	90.9	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.4	6.4	69.1	97.0	1.3	1.2	68.1	97.9	67.3	98.5	66.6	99.2	65.6	100.0		
(5)			67.5	85.7	1.2	1.3	66.6	86.6	65.9	87.4	65.2	88.1	64.3	89.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.3	81.1	81.6	82.3	82.9	83.2	82.9	82.6	82.6	82.2	82.1	82.0	81.7
	DBStd	1.84	1.73	1.78	1.69	1.62	1.76	1.87	1.96	1.97	1.93	1.60	1.40	1.54
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11774	963	884	1002	986	1028	986	1012	1010	967	994	959	983
	CDD65	6299	498	464	537	536	563	536	547	545	517	529	509	518
	CDH74	56479	3998	3910	4812	4942	5412	5144	5200	5198	4721	4565	4307	4270
	CDH80	22744	1382	1415	1905	1997	2248	2201	2322	2293	1997	1807	1611	1566
(14) Wind	WSAvg	2.2	2.1	2.2	2.1	2.2	2.2	2.2	2.3	2.3	2.1	2.0	2.1	2.1
(15) Precipitation	PrecAvg	132.2	14.7	10.1	10.9	10.8	8.8	7.8	7.3	9.8	11.1	11.8	14.1	15.1
	PrecMax	169.4	20.7	24.3	21.1	16.0	14.2	13.3	14.7	29.1	19.4	18.1	21.3	23.1
	PrecMin	96.7	9.1	3.4	3.1	4.8	3.5	5.2	2.1	2.8	3.0	4.8	7.5	6.1
	PrecStd	18.0	3.3	5.1	4.9	2.9	3.3	2.0	3.3	6.1	4.1	4.4	3.4	3.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.0	93.4	93.6	95.0	95.3	95.3	96.5	95.4	95.1	93.6	93.3	93.0
		MCWB	81.2	81.0	81.1	81.2	81.8	81.3	81.2	80.6	81.0	80.9	81.2	82.0
	2%	DB	91.0	91.5	91.8	93.2	93.5	93.5	93.6	93.6	93.3	91.8	91.5	91.3
		MCWB	80.6	80.2	80.4	81.2	81.6	81.4	80.9	80.4	80.9	80.5	80.9	81.2
	5%	DB	89.4	89.7	91.2	91.6	91.8	92.8	93.0	92.9	91.6	91.2	89.8	89.6
		MCWB	80.0	79.6	80.1	80.6	81.2	81.3	80.8	80.3	80.4	80.3	80.3	80.5
	10%	DB	87.6	87.9	89.5	89.8	91.0	91.0	91.1	91.1	89.9	89.4	88.1	87.9
		MCWB	79.8	79.5	79.8	80.2	81.0	80.8	80.2	80.0	80.0	79.8	80.2	80.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.6	82.7	83.1	83.1	83.9	83.7	84.5	82.8	83.5	83.0	82.9	83.4
		MCDB	89.3	90.3	91.1	91.6	92.6	92.2	92.5	91.8	92.0	90.9	90.1	90.4
	2%	WB	81.4	81.3	81.8	82.0	82.7	82.7	82.3	81.8	82.1	81.7	81.7	81.9
		MCDB	88.1	88.6	89.8	90.2	91.2	91.1	91.3	90.8	90.5	89.6	89.0	88.9
	5%	WB	80.5	80.3	80.8	81.2	81.9	81.8	81.2	81.0	81.1	80.8	80.9	81.0
		MCDB	87.2	87.3	88.5	89.2	89.9	90.0	90.0	89.9	89.3	88.5	88.0	87.9
	10%	WB	79.8	79.7	80.0	80.5	81.0	80.9	80.4	80.1	80.2	80.0	80.0	80.1
		MCDB	85.9	86.2	87.1	88.0	88.6	88.8	88.9	88.6	87.9	87.2	86.7	86.5
(35) Mean Daily Temperature Range	5% DB	MDBR	14.1	15.0	15.7	15.8	16.0	16.5	17.2	17.1	16.4	15.7	15.1	14.7
		MCDBR	16.7	16.9	17.7	17.8	18.1	19.3	20.1	20.1	18.9	18.2	16.8	16.5
		MCWBR	7.7	7.4	7.6	7.9	8.3	8.9	9.1	8.9	8.8	8.3	7.9	7.9
	5% WB	MCDBR	15.2	15.5	16.5	16.7	17.0	17.8	18.4	18.2	17.7	16.6	15.9	15.7
		MCWBR	7.5	7.2	7.5	7.9	8.3	8.7	8.9	8.7	8.8	8.2	7.9	8.0
(40) Clear-Sky Solar Irradiance	taub		0.406	0.423	0.427	0.409	0.411	0.409	0.431	0.529	0.578	0.560	0.430	0.402
	taud		2.482	2.410	2.396	2.449	2.450	2.448	2.380	2.092	1.955	1.998	2.402	2.497
	Ebn at Noon		292	288	285	284	277	273	268	248	241	248	283	292
	Edh at Noon		36	39	40	37	36	35	38	52	61	59	39	35
(44) All-Sky Solar Radiation	RadAvg		1407	1483	1577	1595	1563	1545	1577	1575	1465	1452	1484	1454
	RadStd		101	75	69	72	72	82	79	93	147	179	69	63

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	+0.45	+0.27	N/A	+0.67	+0.81	+0.63	N/A	N/A	+234	+234

Nomenclature: See separate page